Robotic Spraying in Specialty Crops

Progress Today, Potential for Nurseries Tomorrow

Examples from orchards, vineyards, tree nuts, and emerging nursery applications

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Nursery Automation Lab

Research
- Precision Agriculture for Nursery Crop Management
- Robotic Solutions for Nursery Crop Production
- Advising graduate and undergraduate students

Teaching
- Guest lectures, future graduate courses

Services
- Scientific reviews, professional society committees, extension activities etc.

Team

Lab members

PI: Chenchen Kang

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Nursery Automation and Robotics Laboratory at the Otis I
joining TSU, I was a postdoctoral scholar in Agricultural ar
He. I earned my Ph.D. from Washington State University ir
M.S. and B.S. in Agricultural Engineering from China Agric

PhD student (2026 -)

(Hiring now)

MS student (2026 -)

(Hiring now)

Spray in nurseries

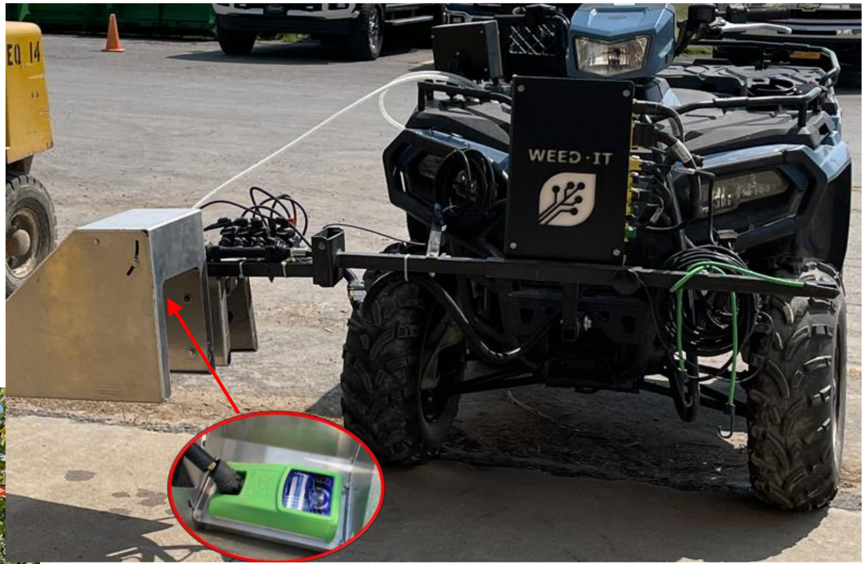
- Diseases, insects, and weeds.
- Critical in southeast region.
- Challenges: drift, contamination, human health concerns.
- Solutions: Precision spraying, robotic spraying.

What Are on the Market (or Under Development)

Precision Spray Technologies

Variable rate sprayer

Robotic sprayer



Precision weed control

Clarifying Terms

Precision sprayer vs. unmanned sprayer vs. robotic sprayer

PS

Precision Sprayer

- Defined by how it sprays
- Improves application accuracy
- May still be operated by a human-driven tractor

Examples: Smart Apply, WEED-IT

UM

Unmanned Sprayer

- Defined by where the operator is
- No operator physically onboard
- May be remotely controlled or autonomous

Examples: spray drones, GUSS

RO

Robotic Sprayer

- Defined by how the system operates
- Combines mobility, sensing, and spray actuation
- Operates with limited human intervention

!

Key distinction

Precision sprayer = smarter spraying. Unmanned sprayer = no operator onboard. Robotic sprayer = sensing, moving, and spraying with autonomy.

Major Categories of Robotic Spraying Systems

Grouped by system architecture

Autonomous tractor + sprayer

A self-driving tractor pulls or carries a conventional or precision sprayer.

Example: Kingman Ag + OnTarget

UGV + sprayer

A compact unmanned ground vehicle carries or pulls a spray unit.

Example: Burro Sprayito

Integrated autonomous sprayer

The vehicle and sprayer are designed as one autonomous machine.

Example: GUSS

Spray drone

An unmanned aerial platform performs spray application from the air.

Examples: DJI Agras, Hylio

Cross-cutting layer: Precision spray modules | OnTarget, Smart Apply, WEED-IT, LiDAR canopy sensing, spot spraying, variable-rate control

Autonomous tractor + sprayer

Kingman Ag

Autonomous tractor

OnTarget sprayer

Electrostatic spray

ONTARGET SPRAY SYSTEMS

Autonomous tractor + sprayer

bluewhite

Tennessee State University

Autonomous tractor + sprayer

bonsai

Affordable autonomy for any off-road job

UGV + sprayer

BURRO

ONTARGET SPRAY SYSTEMS

UGV + sprayer

BURRO

WEED·IT

precision spraying

Tennessee State University

UGV + sprayer

bonsai

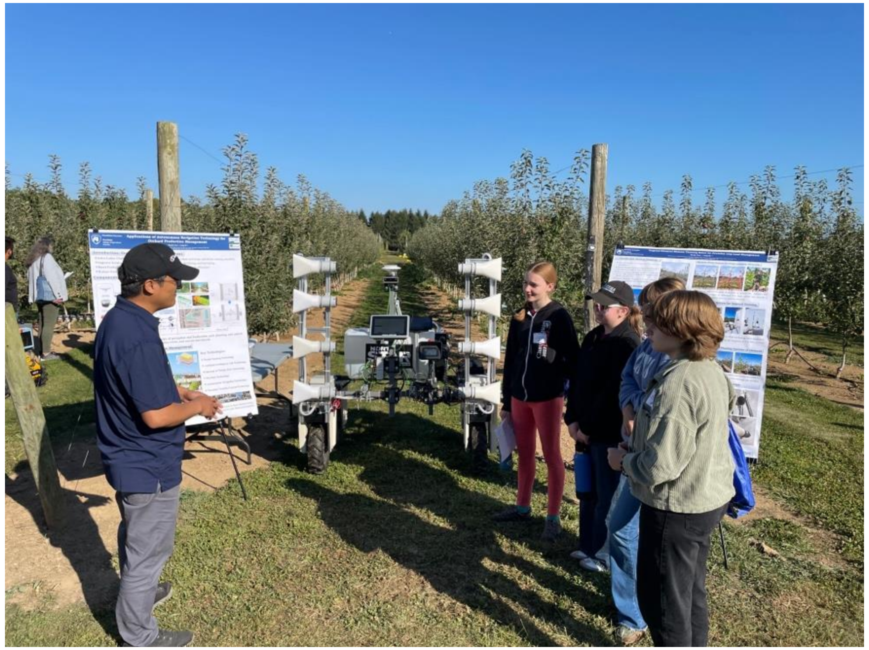
Affordable autonomy for any off-road job

ONTARGET SPRAY SYSTEMS

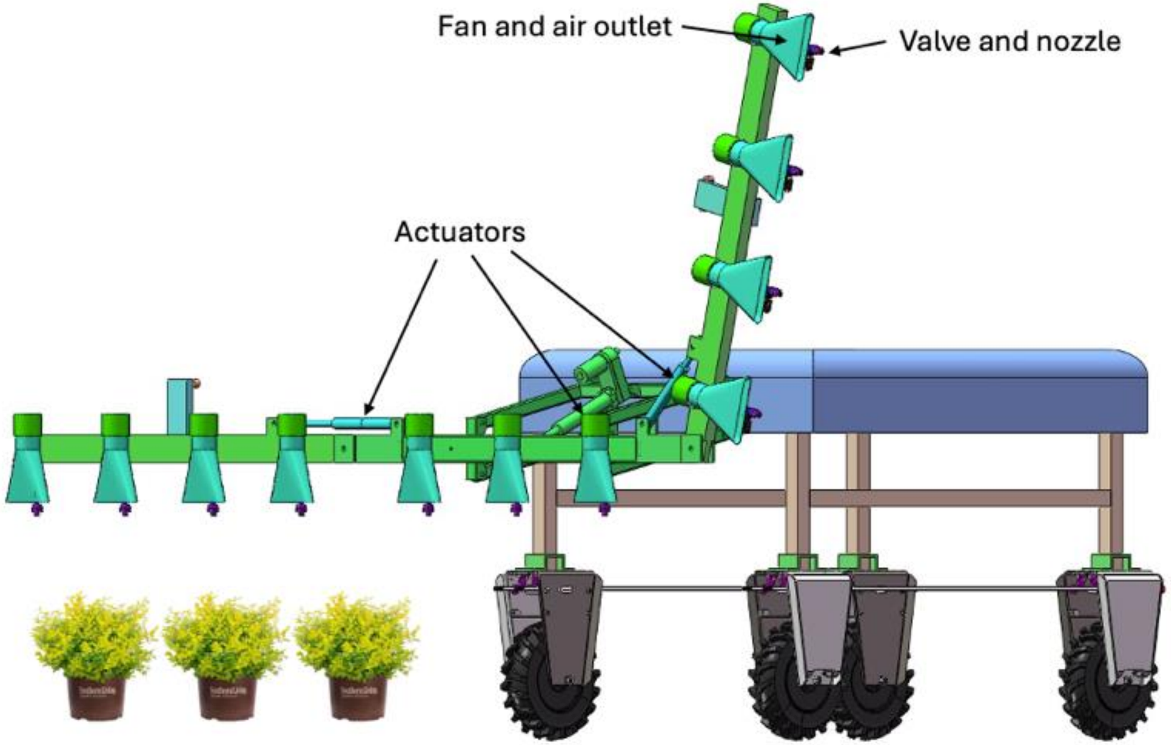
UGV

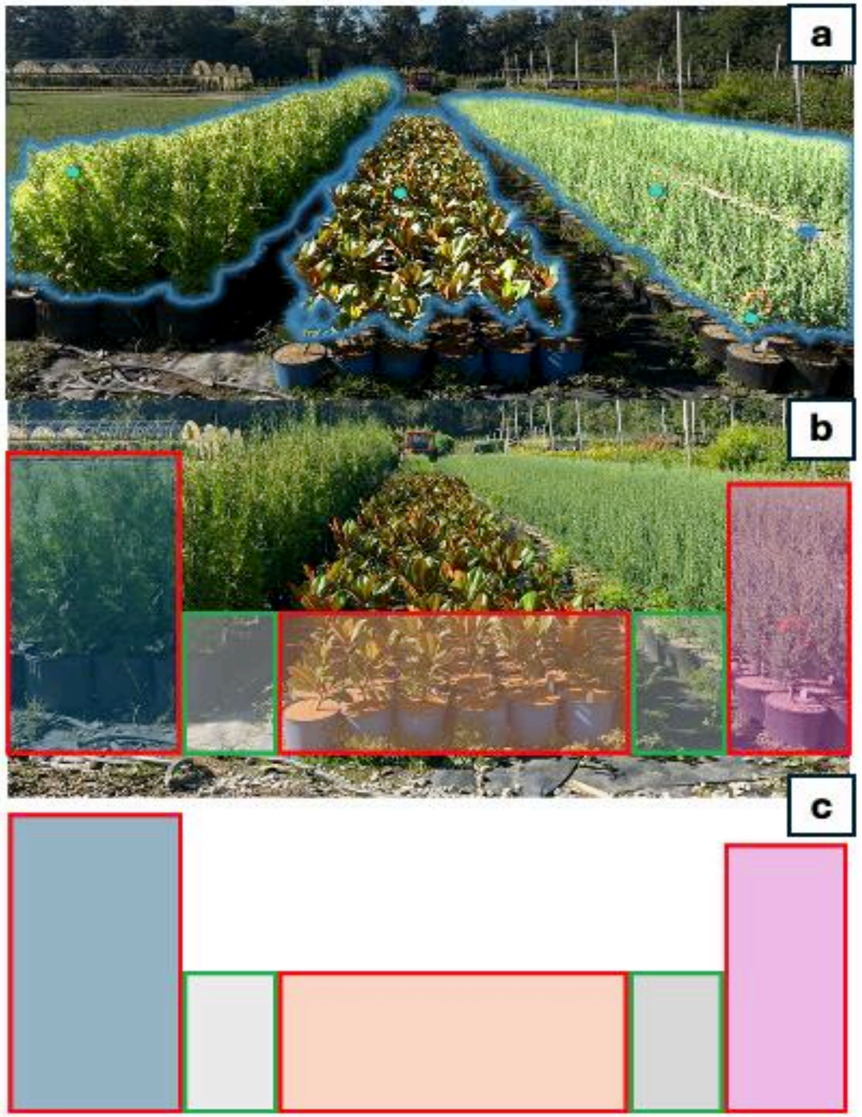
Bonsai Amiga platforms


bonsai



UGV + sprayer





Robotic sprayer for container nursery

UGV + spreader

Broadcast application between rows

Band or targeted application

Robotic Sprayer

Robotic Sprayer

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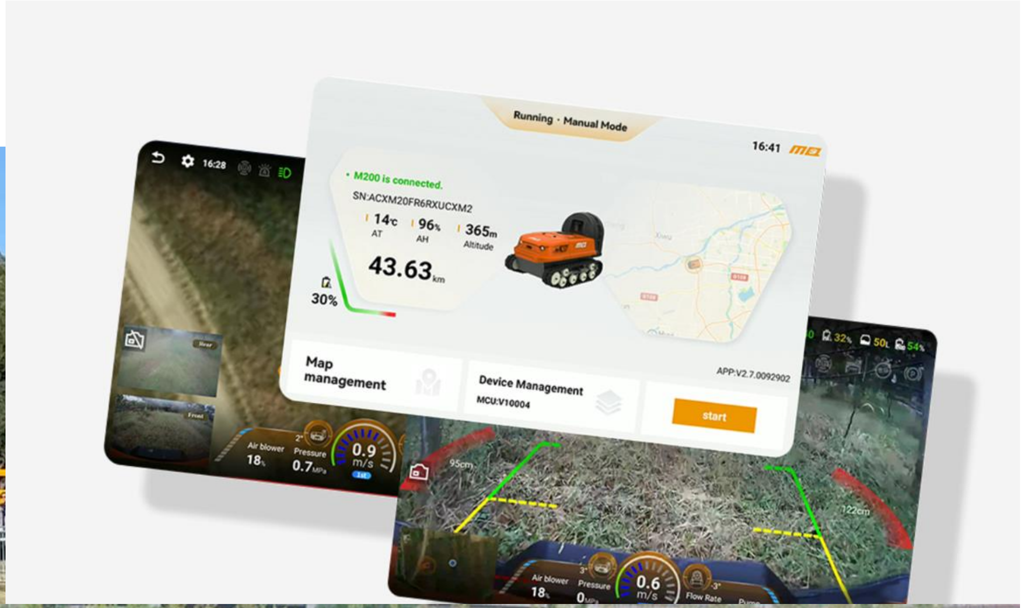
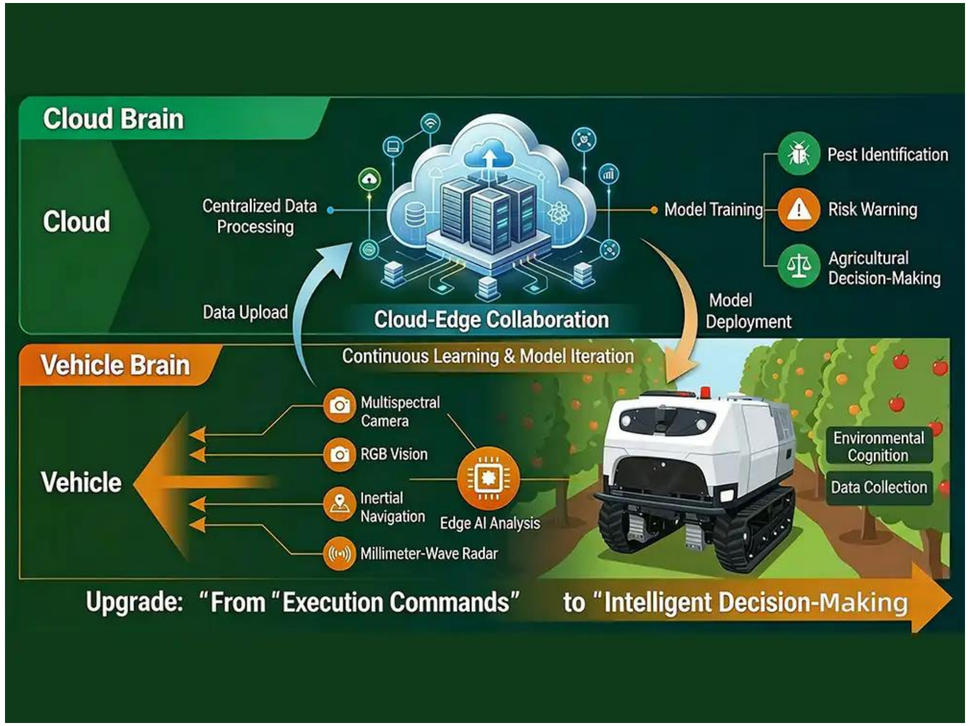


Robotic Sprayer

LJ Tech



Robotic Sprayer

TENNESSEE STATE UNIVERSITY

GUSS
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JOHN DEERE

Drone Sprayer

Drone Sprayer

Drone Sprayer

Drone Sprayer

Summaries

- ❖ A nursery automation lab is established and will work closely with the nursery industry.
- ❖ Currently available robotic spraying technologies for growers.
- ❖ Actively exploring the possibility of tech adoption for nursery growers.

Thank you!

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